Higgs boson decay to two photons Experiment

Third Year Lab Report

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This experiment aims to measure the interaction cross-section for the decay of the Higgs boson into two photons at the Large Hadron Collider at a center-of-mass energy $\sqrt{s}$ of 13 TeV. This experiement has applied a 'hidden box' method at invariant mass of 122-128 GeV and the determined cross-section from this method is 70.0 ± 52.4 fb.

1. INTRODUCTION

The Higgs boson is a theoretical particle proposed by Peter Higgs and serves as the exchange particle for the Higgs field, which explains the origin of mass for particles. In 2012, the ATLAS detector at the Large Hadron Collider confirmed the existence of the Higgs boson through high-energy proton-proton collisions. These collisions disrupt the protons and release high-energy gluons, which predominantly produce the Higgs boson via a virtual top quark loop[1]. Due to the Higgs boson's extremely short lifetime of 1.9×10^{-13} seconds, it cannot be detected directly by the ATLAS detector[2]. Instead, the detector is designed to capture the products of Higgs boson decays. This experiment focuses on the decay of the Higgs boson into two photons, a decay product detectable by the ATLAS detector's ability. We have gathered photon data from the ATLAS detector and conducted various analyses to determine the cross-section of the Higgs-to-two-photon interaction.

1.1. Monte Carlo simulation

Monte Carlo(MC) simulation plays a pivotal role in the analytical processes of this experiment. This method utilizes computational algorithms to simulate the probability distributions of various outcomes in particle interactions and decays[3]. Our study employed Monte Carlo simulations to generate expected signal distributions for Higgs boson decay to two photons.

1.2. Cross-section

In particle physics, the cross-section quantifies the probability of a particular interaction or scattering event when two particles collide. This measure is analogous to the effective target area that one particle presents to another during their encounter[4]. Given the exceedingly small magnitude of the cross-section in particle interactions, it is expressed in the unit of femtobarns (fb), where one femtobarn equals 10^{39} square centimeters. This unit helps quantify and compare the likelihood of various particle collision outcomes within the experimental framework.

2. THEORY AND METHODOLOGY

2.1. invariant mass

Our analysis in the experiment utilizes the invariant mass histogram. Many particles are not directly detectable by the ATLAS detector; hence, the invariant mass is a derived quantity essential for inferring the presence of the original, undetectable particles. Crucially, the invariant mass of a system remains constant across all frames of reference, even in the context of relativistic collisions[5]. This consistency makes it a reliable metric for particle identification. The equation for the invariant mass of the Higgs boson is given by

$$m = \sqrt{2p_{T1}p_{T2}\left(\cosh(\eta_1 - \eta_2) - \cos(\phi_1 - \phi_2)\right)}, \quad (1)$$

where m is the invariant mass in the unit of MeV/c^2, p_{T1} and p_{T2} represent the transverse momentum of two photons produced in the unit of MeV/c which is the component of the particle's momentum perpendicular to the beam axis. ϕ_1 and ϕ_2 represent the azimuthal angles of the particles, which is a dimensionless quantity measured in the plane perpendicular to the beam axis. η_1 and η_2 are the pseudorapidities of the two particles, which is a dimensionless quantity measured for the angle relative to the beam axis. Figure 1 shows the normalized histogram of invariant mass between ATLAS data and Monte Carlo simulation after we extract the variable and perform a calculation with the formula 1.

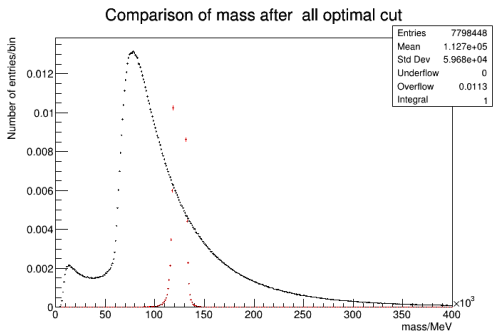

FIG. 1: This figure demonstrate the normalized histogram of invariant mass on data(black) and MC simulation(red)

2.2. Selection criteria

The concept of significance is pivotal in this experiment for assessing the ratio of the signal relative to statistical noise. The significance can be calculated with [6]

$$Z = \frac{N_{\text{signal}}}{\sqrt{N_{\text{background}}}}, \quad (2)$$

where Z is the significance, N_{signal} and $N_{\text{background}}$ is the number of background events. The number of signal events in our analysis is derived from Monte Carlo simulations. In contrast, the number of background events is approximated as the total number of data events recorded, given that the signal is substantially smaller than the background. The maximum significance of the earlier invariant mass histogram is 0.91, indicating that the signal could be easily misinterpreted as statistical noise. This low significance level necessitates robust selection criteria to filter the raw data captured by the ATLAS detector effectively. This

selection process aims to eliminate as many background events as possible while retaining the majority of the signal events. This process can be achieved by investigating the different photon properties detected by the ATLAS detector.
The photons resulting from Higgs boson decays can be distinguished from other background interactions using several measurable properties recorded by the ATLAS detector. The primary distinguishing feature is the photon-tight identification criterion. This method differentiates between genuine photons and background noise caused by electrons, jets, or electronic noise. Additionally, the transverse momentum is another critical variable that needs to be refined in the data. This variable refers to the component of a particle's momentum perpendicular to the beam axis in the ATLAS detector. Further isolating these photons involves analyzing the scalar sum of transverse momentum (ptcone) and transverse energy (Etcone) of all tracks within a cone of radius 0.2 and 0.3, respectively, around the photon's track[7]. This measure helps determine whether the detected photon is isolated, a key characteristic expected in Higgs boson decays to two photons. To enhance the effectiveness of the selection criteria, it is essential to identify the range of each variable that yields maximum significance. This process ensures that the most relevant photon data is retained, maximizing the likelihood of accurately identifying Higgs boson decay events.

2.3. *Calculation for cross section*

The cross section calculation for proton-proton collision in this experiment is given by [4]

$$\sigma = \frac{N_{\text{selected}} - N_{\text{background}}}{\epsilon \cdot \int L\,dt}, \tag{3}$$

where σ is the cross section in unit of fb, N_{selected} is the total number of events in the ATLAS data that pass the final selection cuts in a selected region, $N_{\text{background}}$ is the estimate of the number of background events in the selected region , $\int L\,dt$ is the integrated luminosity with value of 10.064 fb^{-1} which is a measure of the number of proton–proton collisions in ATLAS to which the data sets correspond and ϵ is the efficiency for selecting the signal events which is given by the number of events in Monte Carlo simulation for a selected range of invariant mass divided by the total number of events in Monte Carlo simulation which is 1149.53 in this experiment.

2.4. *Hidden box*

This experiment employs a unique analysis technique involving a "hidden box" approach, which is used to temporarily obscure the signal region to construct a fitting function for the background histogram. This method aims to estimate the background events within the signal region accurately. A reduced chi-square test is used to determine whether the fitting function is valid to approximate the background of data, and the reduced chi-square can be calculated by

$$\chi^2_{\text{red}} = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{\sigma_i^2 \nu}, \tag{4}$$

where n is total number of data points, O_i is the observed value for the i-th data,E_i is the expected value for the i-th data , σ_i is the uncertainty for the i-th data and ν is the degrees of freedom in the analysis. The fitting function will be considered invalid when the reduced chi-square is below 0.5 and above 2.0.

In this experiment, a variety of fitting functions are employed, including exponential(Expo), Gaussian(Gauss), and polynomial functions of second, third, and fourth orders (Poly2, Poly3, Poly4), as well as a composite function that combines an exponential with a second-order polynomial (Poly2+expo). The rationale behind utilizing multiple fitting functions is to diversify the results produced, thereby minimizing systematic errors. The fitting ranges are set to be different. This strategy allows for more robust analysis, as it helps to mitigate biases or anomalies specific to a particular range or model. By expanding the variety of fitting ranges, we enhance the reliability of the results and further decrease systematic uncertainties.

3. RESULTS AND ANALYSIS

3.1. *Selection*

The analysis to identify regions with maximum significance for each variable established the selection criteria summarized in Table I. The maximum significance of invariant mass for this selection set is 3.24, which is large enough to conduct further analysis and calculation.

TABLE I: Selection Criteria

Quantity	**Cut Value[MeV]**
First Photon Transverse Momentum	$53000.0 - 200000.0$
Second Photon Transverse Momentum	$38000.0 - 200000.0$
First Photon Etcone20	$-4200.0 - 4200.0$
Second Photon Etcone20	≤ 3000.0
First Photon ptcone30	≤ 4200.0
Second Photon ptcone30	≤ 3700
First Photon TightID	1
Second Photon TightID	1
Invariant mass	$122000.0 - 128000.0$

This table summarizes the selection criteria used in the analysis. The first and second photon indicate the photon with maximum and second maximum for each quantity.

3.2. *Cross section*

The invariant mass range where the Higgs boson signal is anticipated, specifically between 122 GeV and 128 GeV, as detailed in Table I, was obscured using the hidden box method. This technique is essential for estimating the background within this crucial signal range. Figure 2 presents a histogram utilized in the process to $N_{\text{selected}} - N_{\text{background}}$, which represents the net signal after background subtraction. This method was systematically applied using various fitting functions and ranges to ensure robustness in our background estimation.

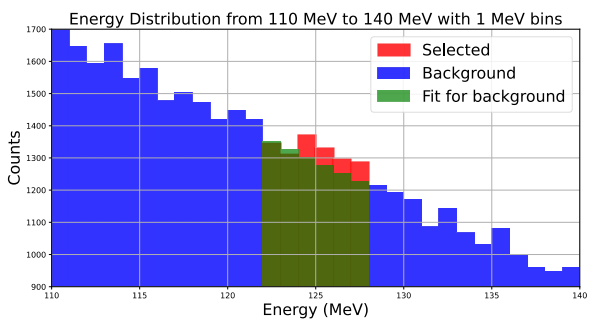

FIG. 2: This figure demonstrates an example of the hidden box method we have used in one function to obtain the value of N_{selected}(Red) - $N_{\text{background}}$(Green)

The cross-section is then calculated by the formula 3 with these values. Table II shows the cross-section calculation conducted for each scenario. After we exclude the value that did not pass the chi-square test, the mean cross-section for Higgs boson decays to two photons is 70.0 fb.

TABLE II: Cross-section

Range(MeV)	Expo	Gauss	Poly2	Ploy2+expo	Poly3	Poly4
100-190	148.43	100.97	75.28	90.01	73.77	41.80
110-180	92.78	83.56	66.89	73.50	71.19	69.92
120-170	42.60	64.75	44.05	35.51	50.79	39.58
110-170	93.53	80.39	69.43	76.84	58.43	63.40
110-160	93.47	64.02	60.99	68.03	81.02	80.96
110-150	86.92	72.58	79.91	83.83	75.50	69.77

This table shows the cross section for each condition. The red values are the scenario that the fitting did not pass the chi square test. The first column is the fitting range for background. The unit of cross section is fb

4. DISCUSSION

4.1. Uncertainty

In this experiment, the total uncertainty in the cross-section measurement comprises three main components: statistical uncertainty, systematic uncertainty, and luminosity uncertainty. Each type of uncertainty stems from different sources and affects the precision and reliability of the experiment's results. Statistical uncertainty arises from the random Poisson fluctuations in data, which is the main contribution of uncertainty. After applying error propagation on formula 3, the cross-section uncertainty due to statistical error is calculated by

$$\Delta\sigma_{stat} = \sigma \times \frac{\sqrt{N_{\text{selected}}}}{N_{\text{selected}} - N_{\text{background}}}, \tag{5}$$

where $\sqrt{N_{\text{s}}}$ is the square root of the number of data events, reflecting the uncertainty in the count of events in the selected region. After incorporating the expression for the cross-section, the formula for statistical uncertainty simplifies to[4]

$$\Delta\sigma_{stat} = \frac{\sqrt{N_{\text{s}}}}{\epsilon \cdot \int L\,dt}, \tag{6}$$

where ϵ is the efficiency, and $\int L\,dt$ is the integrated luminosity. This calculation assumes a negligible variation in N_{s} across different fittings, leading to a consistent statistical uncertainty value of 50.6 fb for all fitting functions. Systematic uncertainty is calculated based on the variation in the cross-section values obtained from different fitting functions. From Table II, the standard deviation of these cross-section values is taken as the systematic uncertainty, which amounts to 13.7 fb. This type of uncertainty accounts for differences in methodological approaches and model dependencies. Luminosity uncertainty arises from errors in measuring the integrated luminosity ($\int L\,dt$) of the experiment. The value of uncertainty can be calculated by replace the $\frac{\sqrt{N_{\text{selected}}}}{N_{\text{selected}} - N_{\text{background}}}$ term in formula 6 by $\frac{\Delta \int L\,dt}{\int L\,dt}$. This replaced term represents the percentage error in the luminosity measurement which is 1.7% in this experiment[7]. The resultant luminosity uncertainty in the cross-section is 1.2 fb. The total uncertainty in the cross-section measurement is derived by combining the individual uncertainties using the root of the sum of squares method, yielding a total uncertainty of 52.4 fb.

4.2. Compare with theoretical result

The theoretical value for the cross-section of the Higgs boson decaying to two photons, calculated to be 127.9 ± 5.73 fb [7], falls within the range of the experimentally obtained value of 70.0 ± 52.4 fb. Considering the experimental uncertainties, this comparison indicates a reasonable agreement between theoretical predictions and experimental results. As the statistical error is high and become main contribution of uncertainty, more selection can be added to reduce this error in future work.

5. CONCLUSIONS

In this experiment, we determined the cross-section of Higgs boson decays to two photons, measuring at 70.0 ± 52.4 fb. This value is consistent within the error margins with the theoretical prediction of 127.9 ± 5.73 fb, which validates the 'hidden box' methods and approaches used in our analysis. Future experiments can build on this work by refining the selection criteria and exploring more sophisticated statistical techniques to reduce uncertainties.

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